



University
of Exeter

Text Mining & NLP

COMM040

Dr. Rodrigo Wilkens

Outline

Recap & Motivation

What is IE?

Why Information Extraction Is Hard

IE as a Pipeline

Beyond Entities

Modern IE

Outline

Recap & Motivation

What is IE?

Why Information Extraction Is Hard

IE as a Pipeline

Beyond Entities

Modern IE

1. Understand the strengths, weaknesses and biases of different NLP methods
2. Know about recent advances and applications of NLP methods, their impact and real-world applications.
3. Use modern NLP techniques to derive insight from unstructured data
4. Communicate the pros and cons and trade-offs of different approaches to text analysis

Previous classes

- Tokenisation, PoS, parsing (dep & const)
- Keywords, gazetteers
- NER, emotions, topic modelling
- Lexical semantics, language models

Outline

Recap & Motivation

What is IE?

Why Information Extraction Is Hard

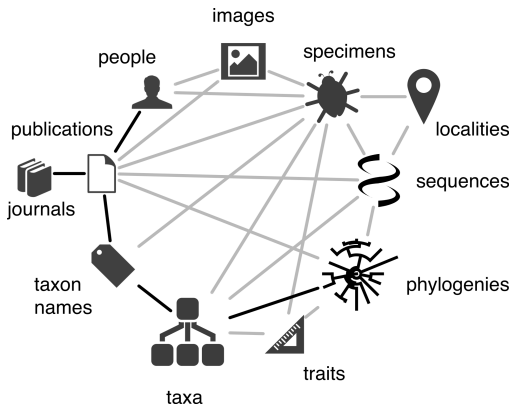
IE as a Pipeline

Beyond Entities

Modern IE

Information Extraction (IE)

- From NLP components to structured knowledge



NLP task vs IE system

- **NLP task**
 - Predicts labels or representations
 - Often evaluated in isolation
- **Information Extraction system**
 - Produces structured outputs
 - Designed for downstream use

Pipeline collage showing tokens → PoS → entities → relations → table (each step faded except the final structure).

IE is schema-driven

In Information Extraction, you must decide **in advance**:

- What fields exist
- What types are allowed
- What relations are valid

→ This is a **design decision**, not a modelling one

Simple database-style schema (field names, types, relations).

From Data to Value

- **Data:** raw symbols and observations
- **Information:** structured, contextualised data
- **Knowledge:** interpreted and connected information
- **Value:** actionable insight and decision-making

DIKW pyramid: Data at the base, Value at the top.

What does “information” mean in IE



University
of Exeter

- Information Extraction aims to extract:

- Entities
- Attributes
- Relations
- Events
- Templates or records

≠ annotation

≠ summarisation

Text → Entities → Relations → Events → Records

Closed IE vs Open IE

- **Closed IE**
 - Fixed schema
 - Task-specific
 - Easier to evaluate
- **Open IE**
 - No predefined schema
 - Predicate-argument triples
 - Exploratory

Outline

Recap & Motivation

What is IE?

Why Information Extraction Is Hard

IE as a Pipeline

Beyond Entities

Modern IE

Why Information Extraction is hard



University
of Exeter

- Ambiguity in natural language
- Implicit and underspecified information
- Cross-sentence dependencies
- Strong domain dependence
- Evaluation

Local accuracy vs global failure

- Individual NLP components may perform well
- Small local errors can invalidate extracted facts
- IE failures are often **systemic**

Pipeline where one small red cross early causes a large failure downstream.

Why rule-based IE still matters

Despite modern ML and LLMs, rules are:

- Highly precise
- Fully interpretable
- Data-efficient
- Easy to debug

Dictionary- and gazetteer-based IE



University
of Exeter

- Match mentions against predefined lists
- Commonly used for:
 - Names
 - Locations
 - Domain-specific terminology
- Limitations: coverage and maintenance

List of names feeding into a text matcher highlighting spans.

Dictionary- and gazetteer-based IE



University
of Exeter

- LIWC

	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU
1																				
2	121 Social				122 Family				123 Friends				124 Humans				125 Affect			
3	126 Posemo				127 Negemo				128 Anx											
4	acquainta	gives	persons	aunt	acquainta	adult	abandon	damn	fume	kind	privileg	supporting	accept	freed	partie	abandon	enrag	maddening	snob	afraid
5	admit	gossip	person's	bro	amigo	adults	abuse	danger	fuming	kiss	prize	supportive	accepta	freely	party	abuse	enrue	madder	sob	alarm
6	admits	gossip	persua	bros	bro	babe	abusi	daring	fun	laidback	problem	support	accepted	freely	passion	abusi	enrue	madder	sob	anguish
7	admitted	grandchild	phone	brother	bi	babies	accept	darin	funn	lame	profit	suprem	accepting	freely	peace	ache	enrue	maddest	sobbing	anxi
8	admitting	granddad	phoning	cousin	bud	baby	accepta	daze	furious	laugh	promis	sure	accepts	freer	perfect	aching	evil	masochis	sobs	apprehens
9	adult	granddad	prais	dad	buddies	bambino	accepted	dear	fury	lazie	protest	surpris	active	freer	play	advers	evictuac	melanchol	solemn	asham
10	adults	grand	private	daughter	buddy	boy	accepting	decay	geek	lazy	protest	suspicio	admir	friend	played	afraid	exhaust	mess	sorrow	avert
11	advice	grandd	provide	es	companion	boy's	accepts	defeat	genero	labilit	protesting	sweet	ador	fun	playful	aggravat	fail	messy	sorry	avoid
12	advic	grandm	public	eyes	companion	boys	achieve	defect	gentle	liber	proud	sweetheart	advantag	funn	playing	aggress	fake	miser	spite	awkward
13	affair	grandpa	question	es	companions	chick	aching	defenc	gentle	liber	puk	sweetie	adventur	genero	plays	agitat	fatal	miss	stammer	confus
14	amigo	grandson	reassur	es	comrad	chick	active	defens	gentlest	lies	punish	sweetly	affection	gentle	pleasant	agoniz	fatig	misses	stank	craz
15	anybod	granny	receiv	es	comrad	chicks	admir	definite	gently	like	radian	sweetness	agree	gentle	pleasur	agony	fault	misses	stank	desperat
16	anyone	group	refus	es	exibit	child	ador	definitely	gigg	like	rage	sweets	agreeab	gentlest	pleasing	alarm	fear	missing	steal	discomfort
17	apolog	grounp	relationship	es	exibit	children	advantag	degrad	giver	likeab	raging	talent	agree	gently	pleasur	alone	feared	mistak	stench	distract
18	argu	gudge	relatives	family	exigif	child's	adventur	delectabl	giving	liked	randic	tantrum	agreeing	gigg	popular	anger	fearful	mock	stink	distraught
19	armies	guest	reple	father	exigif	citizen	advers	delicate	glad	likes	rape	tears	agreement	giver	positiv	angr	fearing	mocked	strain	distress
20	army	guy	reply	folks	exigif	citizen	affection	delicious	glady	liking	raping	teas	agrees	giving	prais	angr	fears	mocked	strange	disturb
21	ask	he	request	grandchild	friend	citizens	afraid	delight	glamor	live	rapist	tehe	alight	glad	precious	annoy	feroc	mocking	stress	doubt
22	asked	hear	respond	granddad	gl	female	aggravat	depress	glamour	LMAO	readiness	temper	amaz	glady	pretie	antagoni	feud	mocks	struggl	dread
23	asking	heard	role	granddad	girlfriend	gentlem	aggress	depriv	gloom	LMAO	ready	temper	amor	glamor	pretty	anxi	fiery	molest	stubborn	dwell

Levin's verb taxonomy

- Proposed by Beth Levin (1993)
- Groups English verbs into classes based on:
 - shared meaning
 - shared syntactic alternations
- Assumption:
 - similar meanings → similar argument structures

Examples of Levin verb classes

- **Break verbs**
 - break, shatter, crack, smash
- **Change of state verbs**
 - open, close, melt, freeze
- **Communication verbs**
 - say, tell, report, announce

What regular expressions add

- Regular expressions enable:
 - Explicit span detection
 - Structural pattern matching
 - Fast, deterministic filtering
- They don't
 - No learning required
 - Highly brittle to variation

Typical regex-based extractions

- Common examples include:
 - Dates and times
 - IDs and reference numbers
 - Prices and measurements
 - Fixed-format expressions

Dates, prices, IDs highlighted in different colours within text.

Regular expressions

- A formal language for pattern matching over strings
- Used to detect spans with specific surface forms
- Widely used in Information Extraction pipelines

Why regex are useful in IE

- Fast and deterministic
- High precision for well-defined patterns
- Useful for:
 - dates
 - identifiers
 - prices
 - measurements

Core regex operators

- + : one or more occurrences
- * : zero or more occurrences
- ? : optional
- [abc] : character class
- () : grouping / capture
- Quantifiers:
 - {m} exactly m times
 - {m,n} between m and n times

Regex in Python

- Python uses the `re` module
- Raw strings avoid escaping issues:
 - `r"\d+"`
- Common functions:
 - `re.search`
 - `re.findall`
 - `re.match`

Designing a regex: step by step

- How to build a regular expression?
 1. define what you want to match
 2. identify invariant parts
 3. add controlled variation
 4. test on real examples

Limitations of regex

- No semantic understanding
- Fragile to paraphrase
- Hard to scale to complex language

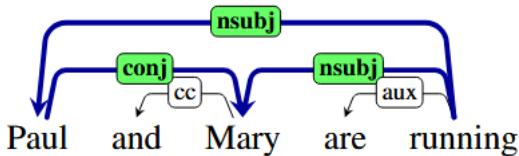
Regex + linguistic context

- Regex can be combined with:
 - token boundaries
 - PoS tags
 - dependency relations

Rule-based IE examples

- Examples of hand-crafted patterns:
 - “X was born in Y”
 - “Company A acquired Company B”
 - Invoice field extraction

```
{word:running}  
  >nsubj ({ } >conj { }=C)  
  >nsubj { }=C
```



Rule-based IE

- **Strengths**

- Very high precision
- Fully explainable
- Effective in constrained domains

- **Weaknesses**

- Low recall
- High engineering cost
- Poor cross-domain scalability

Rules and models are not competitors



University
of Exeter

- Rules and models solve different problems
- Most real IE systems are hybrid
- Design choices matter more than algorithms

Where rules appear in IE pipelines



University
of Exeter

- Rules are commonly used:
 - Before models (filtering, normalisation)
 - After models (validation, correction)
 - Alongside models (ensemble decisions)

Information Extraction as a pipeline problem



University
of Exeter

- Information Extraction is rarely a single model
- IE systems are composed of multiple components
- Design choices shape overall behaviour

Outline

Recap & Motivation

What is IE?

Why Information Extraction Is Hard

IE as a Pipeline

Beyond Entities

Modern IE

A canonical IE pipeline

- Typical components include:
 - Preprocessing and normalisation
 - Entity extraction
 - Relation or event extraction
 - Post-processing and validation



Sequential pipelines

- Components executed one after another
- Example: Named Entity Recognition → Relation Extraction
- Advantages: modularity, interpretability
- Disadvantages: error propagation

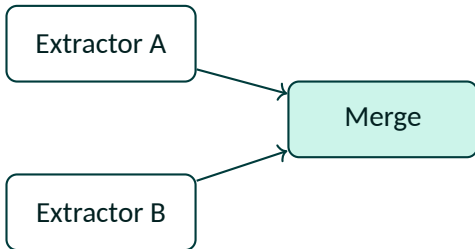
Error propagation

- Early mistakes affect downstream components
- A single wrong entity can invalidate a relation
- Errors compound rather than cancel out

Zoom-in showing one incorrect entity breaking multiple relations.

Parallel pipelines

- Multiple extractors run independently
- Results combined or reconciled later
- Advantages: robustness
- Disadvantages: conflict resolution



The role of syntax in IE pipelines

- Syntactic information may support:
 - Argument identification
 - Relation directionality
 - Clause and scope detection

Dependency tree with subject-object path highlighted.

Designing an IE pipeline

- Key design questions:
 1. What is the target structure?
 2. Which components are necessary?
 3. Where can errors be tolerated?

Outline

Recap & Motivation

What is IE?

Why Information Extraction Is Hard

IE as a Pipeline

Beyond Entities

Modern IE

From entities to relations

- Entities alone rarely capture full meaning
- Facts emerge through **relations**
- IE moves from “things” to “connections”



Beyond triples: richer representation



University
of Exeter

- Predicate–argument triples are a useful abstraction
- Some applications require richer semantics
- Example: FrameNet
 - Frames
 - Frame elements (roles)

What is relation extraction?

- Relation Extraction identifies:
 - A relation type
 - Two or more arguments (entities)
 - Their semantic roles

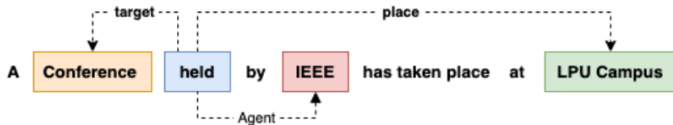
Sentence with two entities connected by a labelled arrow.

Relation extraction in IE pipelines

- Relations are often extracted:
 - After entity recognition
 - Using syntactic or contextual cues
 - With rule-based or statistical components

From relations to events

- Relations capture static facts.
- Events capture:
 - Actions
 - Changes
 - Temporal structure



What is event extraction?

- An event is typically represented by:
 - A trigger (lexical anchor)
 - One or more arguments
 - An event type

Example: *Product launch, acquisition, protest*

Event schemas

- Event extraction requires:
 - Predefined event types
 - Expected argument roles
 - Constraints on participation

Practical simplifications of event extraction

- In practice, systems often:
 - Reduce events to template filling
 - Focus on a small number of event types
 - Ignore cross-sentence events

Limitations of OpenIE

- Shallow semantics
- Difficult evaluation
- Limited usefulness for downstream applications

Closed IE vs Open IE

- Closed IE: precise, schema-driven, task-specific
 - Open IE: flexible, exploratory, shallow
- Choice depends on application goals

Outline

Recap & Motivation

What is IE?

Why Information Extraction Is Hard

IE as a Pipeline

Beyond Entities

Modern IE

Information Extraction with Large Language Models

- Recent LLMs can perform IE without task-specific training
- Extraction becomes a form of controlled generation
- IE shifts from pipelines to prompt design

Prompt → LLM → structured JSON output.

Information Extraction with Large Language Models

- With LLMs, IE is often framed as:
 - controlled generation
 - schema-guided output
 - instruction-following extraction

IE as generation with constraints

- With LLMs, IE is often:
 - Prompted extraction
 - Schema-guided output
 - Structured generation (JSON, tables)

Fine-tuning vs prompting for IE

- Fine-tuning:
 - Task-specific
 - Requires annotated data
- Prompting:
 - Zero- or few-shot
 - Flexible and fast to adapt

Evaluation Challenges with LLM-based IE



University
of Exeter

- Classical IE metrics are often insufficient:
 - partial correctness
 - missing or extra fields
 - hallucinated content
- Evaluation strategies include:
 - schema validation
 - constraint checking
 - human-in-the-loop assessment

LLMs for IE: strengths and weaknesses



University
of Exeter

- **Strengths**

- Long-context reasoning
- Cross-sentence extraction
- Rapid schema adaptation

- **Weaknesses**

- Hallucinations
- Format instability
- Overconfidence

LLMs in real IE systems

- In practice, LLMs are often:
 - Combined with rules and validators
 - Used as proposal generators
 - Placed inside hybrid pipelines

Designing an IE task

- Before choosing methods, ask:
 - What structure do I need?
 - Who will consume the output?
 - What errors are acceptable?

Intended Learning Outcomes

- By the end of this session, students should be able to:
 - **Explain** what distinguishes Information Extraction from isolated NLP tasks
 - **Analyse** why Information Extraction is challenging at the system level
 - **Design** an Information Extraction pipeline given a target schema
 - **Compare** rule-based, ML-based, and LLM-based IE approaches
 - **Justify** IE design choices based on domain and error tolerance

References I

-  Bauer, John et al. (2023). “Semgrex and ssurgeon, searching and manipulating dependency graphs”. In: *Proceedings of the 21st International Workshop on Treebanks and Linguistic Theories (TLT, GURT/SyntaxFest 2023)*, pp. 67–73.
-  Chen, Ruirui et al. (2024). “Is a large language model a good annotator for event extraction?” In: *Proceedings of the AAAI conference on artificial intelligence*. Vol. 38. 16, pp. 17772–17780.
-  Fillmore, Charles J et al. (2003). “FrameNet in action: The case of attaching”. In: *International journal of lexicography* 16.3, pp. 297–332.
-  Jurafsky, Dan and James H. Martin (2023). *Speech and Language Processing. An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*. 3rd edition draft. URL: <https://web.stanford.edu/~jurafsky/slp3/> (visited on 01/05/2026).

References II

-  Kulkarni, Amit and Varun Dogra (2024). “Comprehensive Survey of Event Extraction Methods in Natural Language Processing”. In: *2024 International Conference on Sustainable Communication Networks and Application (ICSCNA)*. IEEE, pp. 925–929.
-  Levin, Beth (1993). “English verb classes and alternations: A preliminary investigation”. In: *U of Chicago P*.
-  Lu, Yaojie et al. (2022). “Unified structure generation for universal information extraction”. In: *arXiv preprint arXiv:2203.12277*.
-  Pennebaker, James W, Martha E Francis, Roger J Booth, et al. (2001). “Linguistic inquiry and word count: LIWC 2001”. In: *Mahway: Lawrence Erlbaum Associates* 71.2001, p. 2001.
-  Xu, Derong et al. (2024). “Large language models for generative information extraction: A survey”. In: *Frontiers of Computer Science* 18.6, p. 186357.

References III



Yu, Zhengqiu et al. (2025). “Evaluating large language models for information extraction from gastroscopy and colonoscopy reports through multi-strategy prompting”. In: *Journal of Biomedical Informatics*, p. 104844.